Predicting Total Enrollment at U.S. Higher Education Institutions

A Multiple Linear Regression Approach Using the 2013 IPEDS Dataset

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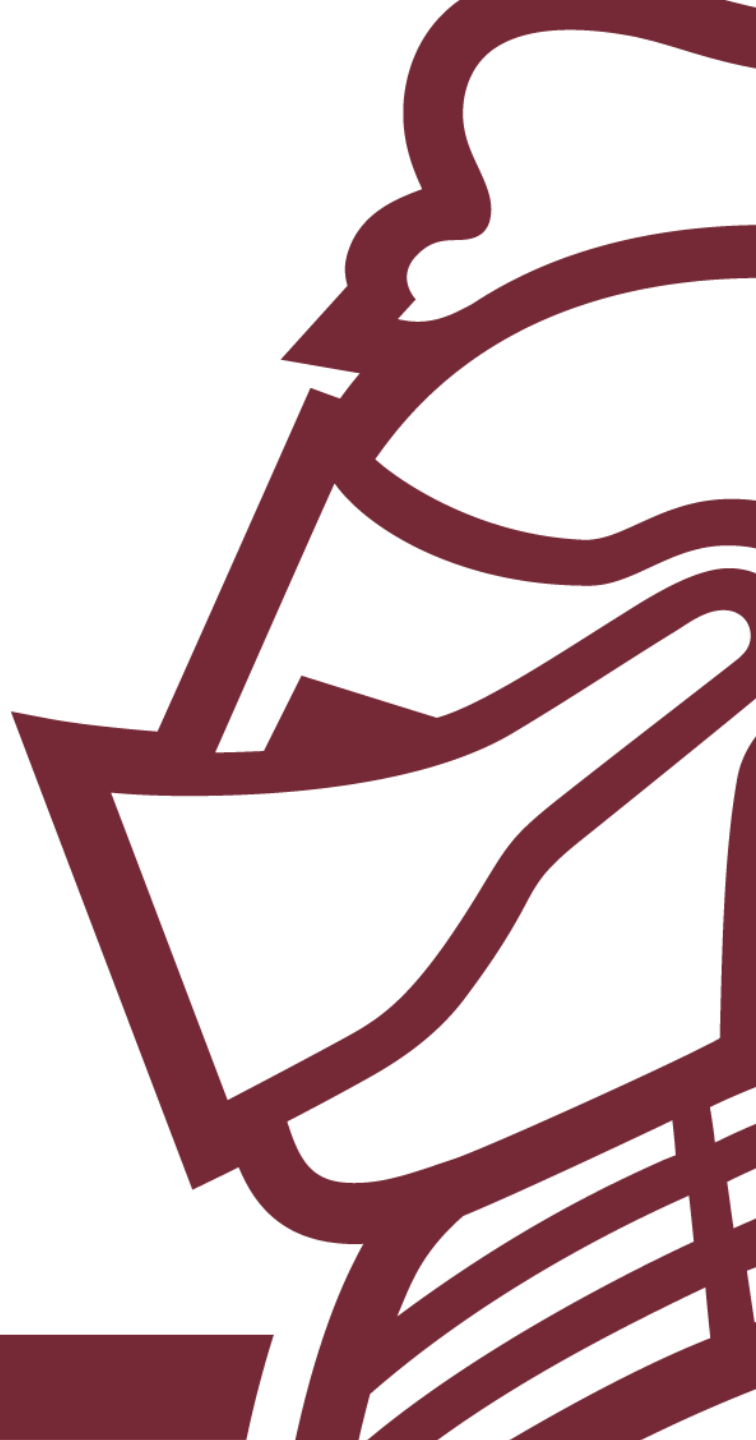
DS160-01 • Final Project



Project Overview

- Predict total enrollment at U.S. higher education institutions from publicly available 2013 data.
- Model type: Multiple Linear Regression (continuous target).
- Dataset: IPEDS 2013, 1,534 institutions, 145 variables.
- Why it matters: enrollment drives revenue, staffing, and capacity, so understanding its drivers is useful for administrators and prospective students alike.

The Dataset: IPEDS 2013

- IPEDS = Integrated Postsecondary Education Data System (NCES).
- Mandatory reporting for every institution in federal student aid programs.
- **Variable naming pattern:**
  - IC2013 → admissions & academic data
  - HD2013 → institutional information (control, location)
  - SFA1213 → financial aid
  - EF2013 → enrollment figures
- **Target: Total enrollment. Range 66 to 77,338 students, median 3,127.**

The Problem & Data Leakage

- **Avoiding data leakage**
  - Target: Total enrollment.
  - Variables like Undergraduate, Graduate, Full-time enrollment are components of the target. Using them yields a fake-perfect model.
  - These variables are excluded from every feature set.
- **Honest predictors used**
  - Numeric: Applicants, % Admitted, Tuition, SAT/ACT scores, Pell %, Graduation rate.
  - Categorical: Control of institution, Highest degree offered.

EDA: Target is Right-Skewed

- **Raw skewness = 2.49**
  - Most schools are small; a long tail of mega-universities (77K+ students) pulls the mean far above the median.
- **log(1 + enrollment) → skew of 0.07**
  - Nearly normal. Errors now interpretable multiplicatively.
- **All models use log target**
  - R^2 reported on log scale; RMSE reported back-transformed to students.

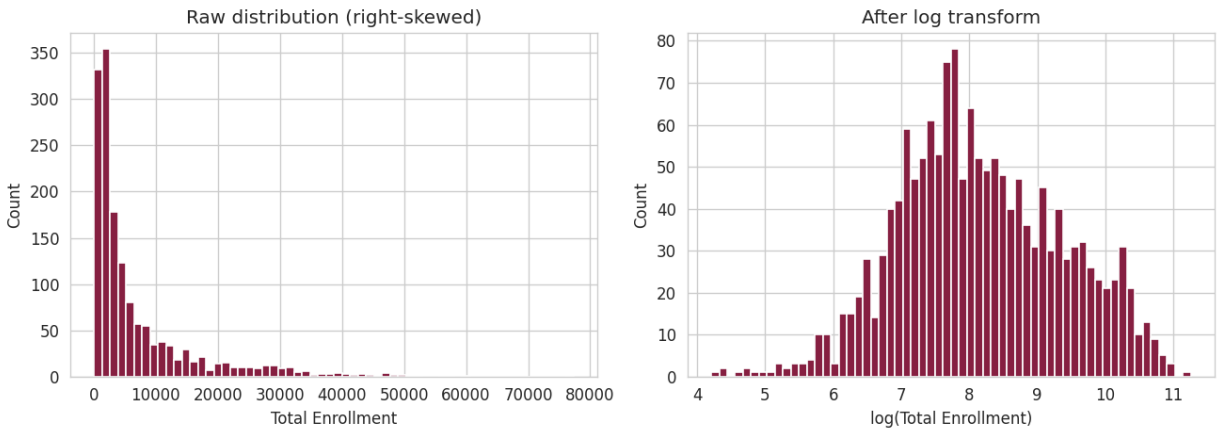


Figure 1: Target distribution before and after log transform

EDA: What Correlates with Enrollment?

- **Strong positive**
  - Admissions (0.73), Applicants (0.67)
- **Moderate positive**
  - SAT Math 75p (0.33), Graduation rate (0.19)
- **Negative**
  - % Financial aid (-0.38), Tuition (-0.28), % Pell (-0.28)
- **Multicollinearity flags**
  - SAT ↔ ACT: r = 0.91 | SAT ↔ Pell: r = -0.75
  - Kept one predictor from each correlated pair in Feature Set B.

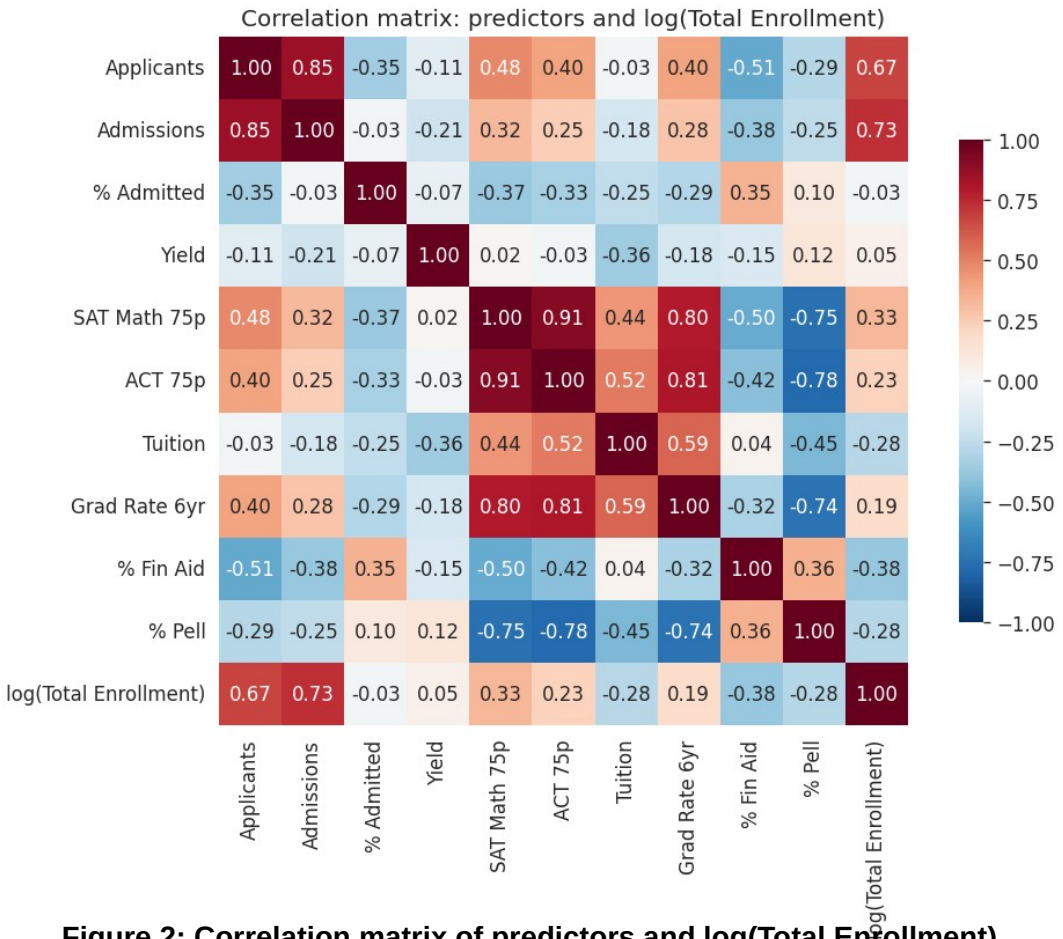


Figure 2: Correlation matrix of predictors and log(Total Enrollment)

Data Preparation

- Dropped 2 rows with missing target.
- Applied log1p transform to the target.
- Complete-case analysis per experiment (no imputation, since SAT is "missing for a reason").
- One-hot encoded categoricals with drop_first=True.
- StandardScaler fit on training set only, applied to both train and test.
- Excluded enrollment-component variables to avoid data leakage.
- Excluded Admissions total (redundant with Applicants × % Admitted).

Experimental Design

- **Two feature sets × two train/test splits = 4 experiments**
  - Feature Set A (4 numeric): Applicants, % Admitted, Tuition, 6-yr Graduation Rate.
  - Feature Set B (A + 4): SAT Math 75p, % Pell, Control of institution, Highest degree offered.
- **Splits**
  - Exp 1: Feature Set A, 80 / 20
  - Exp 2: Feature Set A, 70 / 30
  - Exp 3: Feature Set B, 80 / 20
  - Exp 4: Feature Set B, 70 / 30
- All runs: scikit-learn LinearRegression, random_state=42, standardized features.

Results: Feature Set B Wins

Experiment	Features	Split	R^2 test	RMSE (log)
Exp 1	Set A (4 numeric)	80 / 20	0.588	0.698
Exp 2	Set A (4 numeric)	70 / 30	0.613	0.693
Exp 3	Set B (6 num + 2 cat)	80 / 20	0.776	0.538
Exp 4	**Set B (6 num + 2 cat) ← best**	**70 / 30**	**0.780**	**0.538**

Key takeaways

- Adding 2 categorical variables jumps test R^2 from ~0.59 to ~0.78.
- 80/20 vs 70/30 split barely matters (< 0.01 R^2 difference), so the result holds up.
- Best model MAE ≈ 3,700 students on the raw student scale.

Best Model: Predicted vs. Actual

- **Top standardized coefficients (Exp 4)**
  - Doctoral (research + practice): +0.59
  - Applicants total: +0.44
  - Control = Public: +0.38
  - Doctoral (research/scholarship): +0.35
  - Master's degree: +0.24
  - SAT Math 75p: +0.15
- **Institutional type matters most**
  - Residuals show a mild fan. The model under-predicts the largest universities.

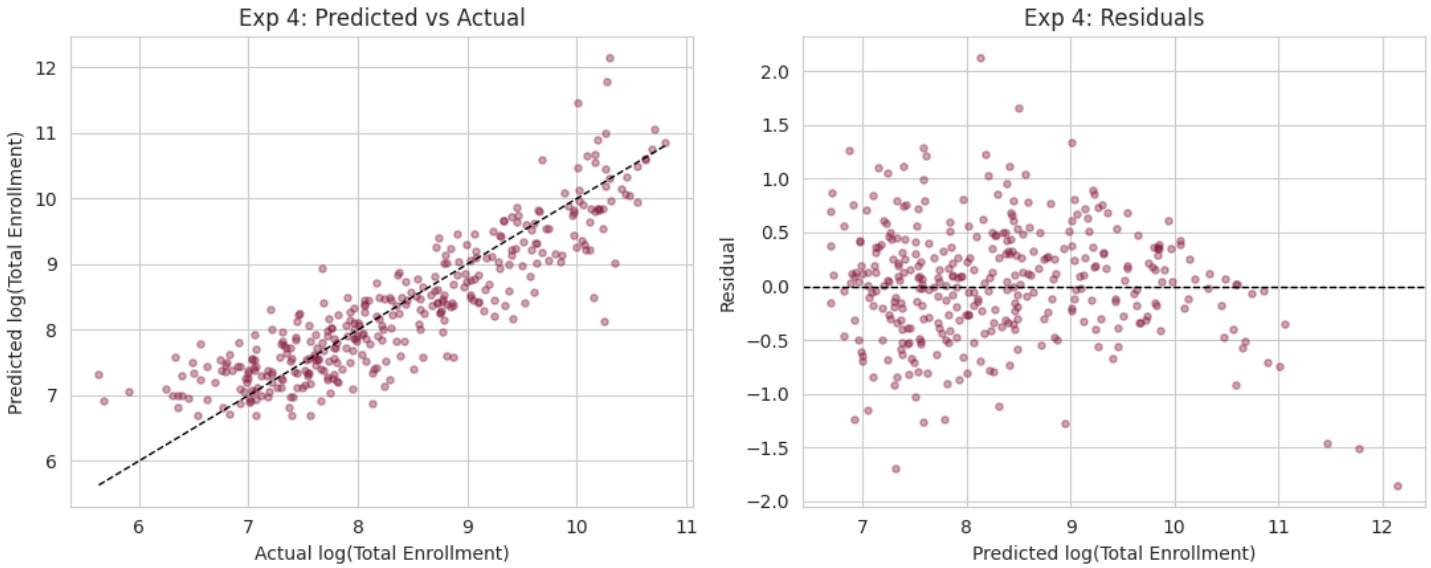


Figure 3: Predicted vs. Actual and Residuals (best model)

Conclusion & Future Work

- **What we learned**
  - Simple OLS explains ~78% of variance in log(enrollment) on held-out data.
  - Institutional type (degree + control) matters more than any single numeric metric.
  - Log transform + avoiding data leakage were the two biggest wins.
- **Limitations**
  - Under-predicts the largest universities; mild heteroscedasticity.
  - Complete-case analysis means results generalize only to SAT-reporting institutions.
- **Future work**
  - Fit a Random Forest / Gradient Boosting regressor and compare.
  - Use multi-year IPEDS data to model enrollment growth rather than level.